

BOOKSWAGON E-Commerce Product Review and Rating Management System

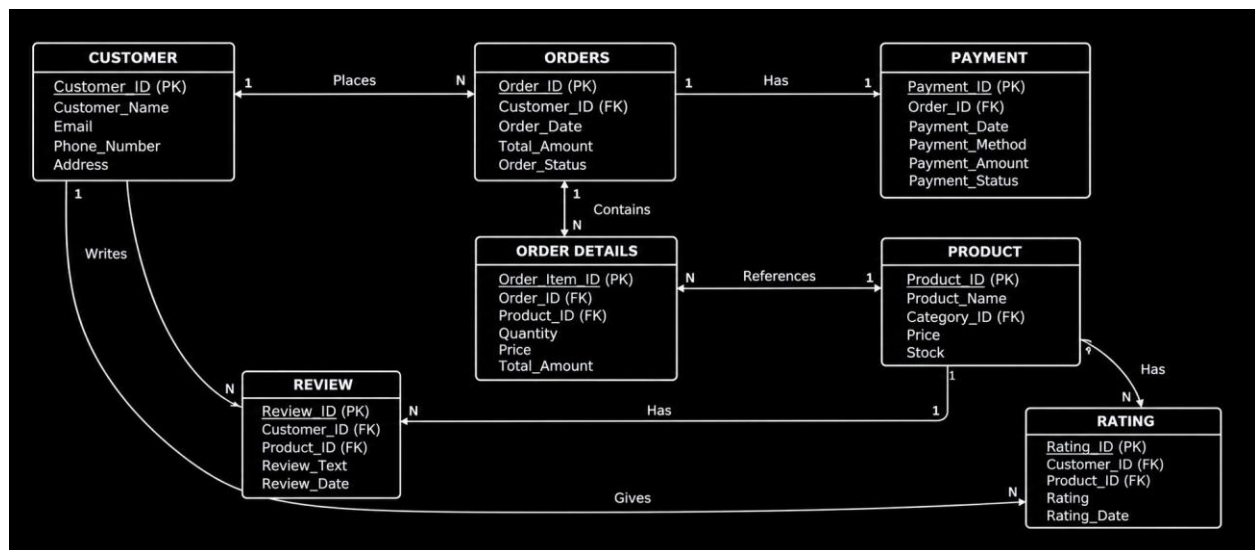
1.1. REVIEW TABLE

Attribute	Data Type	Constraint
Review_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Review_Text	VARCHAR2(500)	NOT NULL
Review_Date	DATE	NOT NULL

1.2. RATING TABLE

Attribute	Data Type	Constraint
Rating_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Rating	NUMBER (1)	NOT NULL
Rating_Date	DATE	NOT NULL

2. TABLE RELATIONSHIPS



3. TABLE CREATION

3.1. CREATE REVIEW TABLE

```
SQL> CREATE TABLE BookswagonReview (  
2     Review_ID NUMBER PRIMARY KEY,  
3     Customer_ID NUMBER REFERENCES BookswagonCustomer(Customer_ID),  
4     Product_ID NUMBER REFERENCES BookswagonProduct(Product_ID),  
5     Review_Text VARCHAR2(500),  
6     Review_Date DATE  
7 );
```

Table created.

3.2. CREATE RATING TABLE

```
SQL> CREATE TABLE BookswagonRating (  
2     Rating_ID NUMBER PRIMARY KEY,  
3     Customer_ID NUMBER REFERENCES BookswagonCustomer(Customer_ID),  
4     Product_ID NUMBER REFERENCES BookswagonProduct(Product_ID),  
5     Rating NUMBER(1) CHECK (Rating BETWEEN 1 AND 5),  
6     Rating_Date DATE  
7 );
```

Table created.

```
SQL> DESC BookswagonReview;
```

Name	Null?	Type
REVIEW_ID	NOT NULL	NUMBER
CUSTOMER_ID		NUMBER
PRODUCT_ID		NUMBER
REVIEW_TEXT		VARCHAR2(500)
REVIEW_DATE		DATE

4. REVIEW DATA INSERTION

```
SQL> INSERT INTO BookswagonReview VALUES
  2 (1, 1, 101, 'Excellent book and very inspiring.', TO_DATE('25-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (2, 2, 102, 'Very useful book for building good habits.', TO_DATE('24-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (3, 3, 103, 'Good book about personal finance.', TO_DATE('24-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (4, 4, 104, 'Very inspiring and interesting biography.', TO_DATE('25-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (5, 5, 108, 'A useful book for motivation.', TO_DATE('26-09-2026','DD-MM-YYYY'));

1 row created.
```

```
SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (6, 6, 107, 'Simple and meaningful book.', TO_DATE('26-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (7, 7, 109, 'Interesting and helpful content.', TO_DATE('27-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (8, 8, 111, 'Amazing story and enjoyable to read.', TO_DATE('27-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (9, 9, 112, 'Good motivational book.', TO_DATE('28-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO BookswagonReview VALUES
  2 (10, 10, 113, 'Very interesting and informative.', TO_DATE('28-09-2026','DD-MM-YYYY'));

1 row created.
```

```
SQL> COMMIT;

Commit complete.
```

```
SQL> SELECT * FROM BookswagonReview;
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
1 1 101
```

```
Excellent book and very inspiring.
```

```
25-SEP-26
```

```
2 2 102
```

```
Very useful book for building good habits.
```

```
24-SEP-26
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
3 3 103
```

```
Good book about personal finance.
```

```
24-SEP-26
```

```
4 4 104
```

```
Very inspiring and interesting biography.
```

```

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
25-SEP-26

          5          5          108
A useful book for motivation.
26-SEP-26

          6          6          107

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
Simple and meaningful book.
26-SEP-26

          7          7          109
Interesting and helpful content.
27-SEP-26

```

```

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
          8          8          111
Amazing story and enjoyable to read.
27-SEP-26

          9          9          112
Good motivational book.
28-SEP-26

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----

          10         10          113
Very interesting and informative.
28-SEP-26

10 rows selected.

```

5. RATING DATA INSERTION

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (1, 1, 101, 5, TO_DATE('25-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (2, 2, 102, 5, TO_DATE('24-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (3, 3, 103, 4, TO_DATE('24-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (4, 4, 104, 5, TO_DATE('25-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (5, 5, 108, 4, TO_DATE('26-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (6, 6, 107, 5, TO_DATE('26-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (7, 7, 109, 4, TO_DATE('27-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (8, 8, 111, 5, TO_DATE('27-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (9, 9, 112, 4, TO_DATE('28-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO BookswagonRating VALUES
  2 (10, 10, 113, 5, TO_DATE('28-09-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL>
SQL> COMMIT;

Commit complete.
```

6. RETRIEVE PRODUCT REVIEW DETAILS

```
SQL> SELECT
  2     r.Review_ID,
  3     c.Customer_ID,
  4     c.Customer_Name,
  5     p.Product_ID,
  6     p.Product_Name,
  7     r.Review_Text,
  8     r.Review_Date
  9 FROM BookswagonReview r
 10 JOIN BookswagonCustomer c
 11 ON r.Customer_ID = c.Customer_ID
 12 JOIN BookswagonProduct p
 13 ON r.Product_ID = p.Product_ID
 14 ORDER BY r.Review_Date DESC;
```

REVIEW_ID	CUSTOMER_NAME	PRODUCT_NAME	REVIEW_TEXT	REVIEW_DATE
9	Vignesh M	The Monk Who Sold His Ferrari	Good motivational book.	28-SEP-26
10	Swetha Priya	The Secret	Very interesting and informative.	28-SEP-26
8	Keerthana R	The Hobbit	Amazing story and enjoyable to read.	27-SEP-26
7	Rahul Kumar	The Power of Your Subconscious Mind	Interesting and helpful content.	27-SEP-26
5	Karthik S	Think and Grow Rich	A useful book for motivation.	26-SEP-26
6	Harini Devi	Ikigai	Simple and meaningful book.	26-SEP-26
1	Nandhini Sri	The Alchemist	Excellent book and very inspiring.	25-SEP-26
4	Divya Raj	Wings of Fire	Very inspiring and interesting biography.	25-SEP-26
3	Arun Kumar	The Psychology of Money	Good book about personal finance.	24-SEP-26
2	Priya Kumar	Atomic Habits	Very useful book for building good habits.	24-SEP-26

10 rows selected.

7.CALCULATE AVERAGE PRODUCT RATINGS

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     ROUND(AVG(r.Rating), 2) AS Average_Rating
  5 FROM BookswagonRating r
  6 JOIN BookswagonProduct p
  7 ON r.Product_ID = p.Product_ID
  8 GROUP BY p.Product_ID, p.Product_Name
  9 ORDER BY Average_Rating DESC;

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      107
Ikigai              5

      104
Wings of Fire      5

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      102
Atomic Habits      5

      113
The Secret

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
          5

      101
The Alchemist
          5

      111

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
The Hobbit
          5

      108
Think and Grow Rich
          4

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      112
The Monk Who Sold His Ferrari
          4

      109
The Power of Your Subconscious Mind
          4

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      103
The Psychology of Money
          4

10 rows selected.

```

8.IDENTIFY HIGHLY RATED PRODUCTS

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     ROUND(AVG(r.Rating), 2) AS Average_Rating
  5 FROM BookswagonRating r
  6 JOIN BookswagonProduct p
  7 ON r.Product_ID = p.Product_ID
  8 GROUP BY p.Product_ID, p.Product_Name
  9 HAVING AVG(r.Rating) >= 4
 10 ORDER BY Average_Rating DESC;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      107
Ikigai          5

      104
Wings of Fire   5

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

      102
Atomic Habits   5

      113
The Secret
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
                    5

          101
The Alchemist
          5

          111

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
The Hobbit
          5

          108
Think and Grow Rich
          4

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
          112
The Monk Who Sold His Ferrari
          4

          109
The Power of Your Subconscious Mind
          4

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----

          103
The Psychology of Money
          4

10 rows selected.

```

9. PRODUCT RATING REPORTS

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     COUNT(r.Rating_ID) AS Total_Ratings,
  5     ROUND(AVG(r.Rating), 2) AS Average_Rating,
  6     MAX(r.Rating) AS Highest_Rating,
  7     MIN(r.Rating) AS Lowest_Rating
  8 FROM BookswagonProduct p
  9 JOIN BookswagonRating r
10 ON p.Product_ID = r.Product_ID
11 GROUP BY p.Product_ID, p.Product_Name
12 ORDER BY Average_Rating DESC;

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS  AVERAGE_RATING  HIGHEST_RATING  LOWEST_RATING
-----
              1              5              5              5

      101
The Alchemist
      1              5              5              5

      111

PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS  AVERAGE_RATING  HIGHEST_RATING  LOWEST_RATING
-----
The Hobbit
      1              5              5              5

      108
Think and Grow Rich
      1              4              4              4

```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS AVERAGE_RATING HIGHEST_RATING LOWEST_RATING
-----
```

```
112
The Monk Who Sold His Ferrari
1 4 4 4
```

```
109
The Power of Your Subconscious Mind
1 4 4 4
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS AVERAGE_RATING HIGHEST_RATING LOWEST_RATING
-----
```

```
103
The Psychology of Money
1 4 4 4
```

10 rows selected.